

## **EDA Report:**

# **Beyond Accuracy: A Clinical Validation Approach to Cardiovascular Disease Prediction Using Machine Learning**

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*DSE6311 Capstone*

Given the trade-off of sensitivity and specificity, how should model performance be evaluated to assess clinically appropriate risk in the context of cardiovascular disease?

Model optimization and validation will balance tradeoff of false negatives and false positives, and the final model may not be the most accurate model according to accuracy/ROC-AUC metrics alone.

A tuned ensemble method such as Random Forest or XGBoost optimized for sensitivity, will outperform Logistic Regression on recall and ROC-AUC. The optimal model will be defined as the sensitivity-specificity combination that best aligns with clinical risk thresholds.

## **Methods:**

For this project The UCI Heart Disease Dataset was initially loaded directly from a public GitHub repository. The dataset was loaded into Google Colab using the `pandas read_csv` function from Python. The dataset contains 303 patient records comprised of 14 total columns. There are 13 features and one binary target variable. The binary target variable indicates the presence of cardiovascular disease(1), or absence of cardiovascular disease (0). Initial inspections of the data showed that there are no missing values across any columns. Further investigation into the data raised concerns about its integrity.

Upon examining the unique values of categorical variables, two anomalies were detected. The variable “ca”, according to original UCI documentation, should contain values 0-3, representing the number of major vessels colored by fluoroscopy, and a value of 4, which is not present in the original documentation. The variable “thal”, which represents thalassemia type, should contain values 3, 6, and 7. This variable was encoded as 1, 2, 3, and 0 and does not correspond with any documented category. Examination of the UCI repository site confirms that both “ca” and “thal” have missing values in the original dataset. This suggests that the GitHub version replaced those missing values with placeholder integers. This practice could hide data quality and potentially introduce errors into the analysis process.

I decided to switch to the official UCI Machine Learning Repository as the data source for reasons of integrity, authenticity, and transparency. The data was accessed by using the `ucimlrepo` Python package with dataset ID 45. The dataset is officially cited as: Janosi, A., Steinbrunn, W., Pfisterer, M., & Detrano, R. (1989). Heart Disease [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C52P4X>.

The official dataset confirmed the expected missing values: “ca” contained 4 missing values (1.32%), and “thal” contained 2 missing values (0.66%). All other variables were complete. The target variable in the original dataset was listed as “num”, and contained 5 levels of disease (0 = no disease, 1 – 4 = varying degrees of disease present). To create a binary target for this project, severity levels 1–4 were collapsed into a single value of 1, representing disease present at any degree. This is consistent with how the Cleveland database has been published in machine learning research. Of the 13 features, 5 are continuous and 8 are categorically encoded as

integers in the raw database. One variable, “oldpeak”, is stored as a float. The variables “ca” and “thal” are also stored as floats due to the missing values.

After inspection of the original dataset, additional encoding differences were found compared to the GitHub version. The “cp” variable used values 1-4, rather than 0-3 and “slope” uses 1-3 rather than 0-2. The “thal” variable has original encoding as 3, 6, 7 for normal, fixed defect, and reversible defect, respectively. The GitHub data dictionary also did not account for these changes. These differences further justify switching the official source.

## **Results:**

Descriptive statistics were evaluated for all continuous and categorical variables.

### ***Continuous:***

The five continuous variables evaluated were age, resting blood pressure, cholesterol, maximum heart rate, and ST depression (“oldpeak”). (see Table 1 and Figure 1.)

Age was the most normally distributed continuous variable with a mean of 54.4 years, a range of 29-77, and mild skewness of 0.21, indicating nearly symmetric distribution. Patient population skews middle to older age, consistent with cardiovascular disease profiles. (CDC, Heart Disease Prevalence, 2023)

Resting blood pressure distribution is reasonably normal, but the upper range is consistent with hypertension. Values averaged 131.69 mmHG with a mild right skew of 0.71 and a maximum value of 200 mmHG. Outliers will be examined during preprocessing.

Cholesterol showed the most concerning distribution of all the continuous variables. A mean of 246.69 mg/dl, right skew of 1.14, kurtosis of 4.49 were found. The kurtosis value indicates heavy tails and the possible presence of outliers. The maximum value of 564 mg/dl seems unreasonable and will be noted for further evaluation in preprocessing.

Maximum heart rate achieved during exercise averaged 149.61 bpm with a range of 71-202, and a kurtosis of -0.05. A mild left skew of -0.54 suggests a small concentration of higher heart rate values.

ST depression induced by exercise, represented by “oldpeak”, showed the strongest right skew at 1.27 with kurtosis of 1.58. Most patients clustered near zero, with a long right tail extending to 6.2. This variable may require transformation during preprocessing. Cholesterol and oldpeak will be prioritized for outlier review and possible transformation in further analysis.

Boxplot analysis confirmed outlier patterns suggested by the skewness in previous statistics. Cholesterol showed the most extreme outlier at 564 mg/dl, sitting well above the upper whisker. Oldpeak showed several outliers as well, ranging from 4.0 to 6.2. Resting blood pressure showed

multiple outliers between 170 and 200 mmHG, which are clinically likely in hypertensive patients and unlikely to be data entry errors. These findings will influence outlier treatment during preprocessing (see Figure 2).

The target variable distribution showed a mild class imbalance with 164 patients without CVD (54.1%), and 139 with CVD (45.9%). This finding is more balanced than anticipated and unlikely to severely distort model performance (see Figure 3). Among the continuous variables, maximum heart rate and ST depression showed the clearest separation between diagnostic classes. Cholesterol and resting blood pressure showed overlap, suggesting weaker predictive value (see Figure 4).

### ***Categorical:***

Evaluation of the categorical variables revealed some clinically meaningful distributions. The dataset was noticeably male-dominated at 68% (n=206), and female patients at 32% (n=97). The gender imbalance is an important demographic limitation to acknowledge as cardiovascular disease presentation and risk factors can differ between genders.

*\*\*\*Note on gender imbalance: Given the time period of data collection (late 1980's), a known period when cardiovascular disease was widely considered a male-dominated condition, this imbalance is not surprising. It was not until the mid 1980's that researchers began examining how heart disease may affect women(Framingham Heart Disease Study), and at the time, research in men was the gold standard(American Heart Association).*

*The historical under-representation of women in CVD research has contributed to an incomplete understanding of symptomology and presentation in women(Gauci, et al, 2022). Women are more likely to experience under-represented symptoms leading to potential delays in diagnosis and treatment (Okumura and Murohara, 2025). This means that female patients in the late 1980's may have been less likely to be referred for a diagnostic workup that could have led to inclusion in this dataset. The gender imbalance is a direct consequence of the dataset's age and is an important demographic limitation. In the further analysis of this project, models trained on this data should be interpreted with caution regarding applicability to female patients. \*\*\**

Asymptomatic chest pain was reported by 47.5% patients, nearly half. This is notable as asymptomatic presentation is a known risk factor for missed cardiovascular disease diagnosis. Fasting blood sugar was elevated in 14.9% of patients. Resting ECG results showed a near even split between normal findings and left ventricular hypertrophy, 49.8% and 48.8%, respectively. Exercise induced angina was present in 32.7% of patients, representing a significant proportion suggesting clinical relevance as a marker of coronary artery disease. There was near even distribution for peak exercise ST segment. Upsloping at 46.9%, flat with 46.2%, and 6.9% showing downsloping pattern. The majority of patients had no major blocked vessels at 58.1% with decreasing proportions at higher vessel counts. Thalassemia type was normal in 54.8% of

patients, reversible defect present in 38.6%, and fixed in 5.9% of patients. Missing values were confirmed in “ca” and “thal”, consistent with original UCI documentation. See Table 2.

When evaluating categorical variables according to CVD diagnosis there were some differences in predictive strength among features. The number of major vessels with blockage showed the most association with CVD prevalence, increasing across vessel count from 0 to 3. Exercise induced angina also had a strong CVD prevalence, with over 75% of patients reporting exercise angina having a positive CVD diagnosis. Asymptomatic chest pain - meaning no noticeable pain is felt, but diagnostic testing such as ECG, imaging or stress tests, confirmed heart disease – showed a high proportion of CVD present. This is consistent with published literature on silent ischemia (Theofilis et al, 2024). These findings will inform feature importance analysis during model development (see Figure 5).

The correlation heatmap confirmed that no single feature dominates CVD prediction (see Figure 6). Thalach showed the strongest negative correlation of -0.42. meaning higher maximum heart rates are associated with lower CVD prevalence. Thal, ca, and exang showed the highest positive correlations with target variable. As thalassemia severity, vessel blockage, and exercise induced angina increase, so does CVD presence. Of the predictor variables, oldpeak and slope showed the highest inter-predictor correlation at 0.58. Both of these variables measure aspects of ST segment behavior and will be monitored for multicollinearity during modeling.

## **Discussion:**

Key takeaways were found during the exploratory data analysis stage. Class imbalance, as noted in Figure 3 , was not as severe as initially anticipated. There are some variables with more extreme outliers that will need to be monitored further (see Figure 2), cholesterol, oldpeak, and resting blood pressure being the most notable. A heatmap of variable correlation (see Figure 6) showed one inter-variable collinearity relationship, oldpeak and slope, that should also be monitored during further analysis. PCA was initially chosen as a dimensionality reduction step to address multicollinearity, but heatmap findings suggest this may not be as critical for future modeling purposes. Plots and graphs of continuous and categorical variables showed that thalach, oldpeak, ca, thal, restecg, and exang emerged as potentially strong predictors based on their visual representations (see Figures 4 and 5).

## **Next Steps:**

Next steps into the preprocessing and feature engineering stage will focus on handling missing values, by dropping or imputation, that appeared in ca and thal variables, treatment of outliers through transformation or removal, and exploring SMOTE for class imbalance, if needed. Variables will be scaled, standardized, and transformed based on skewness findings from EDA, notably cholesterol and oldpeak. The dataset will be split into training and test sets for modeling.

PCA will still be assessed for dimensionality reduction, though the correlation analysis suggests it may be less critical than originally proposed. At this time, specific feature engineering has not been identified and will be driven by preprocessing findings. The original analysis plan has been refined based on EDA findings, particularly regarding class imbalance and multicollinearity present in the data.

**Technical Details:**

Programming language used will be Python, used in Google Colab.

Additional resources used within Python will be scikit-learn, pandas, NumPy, matplotlib, seaborn, XGBoost.

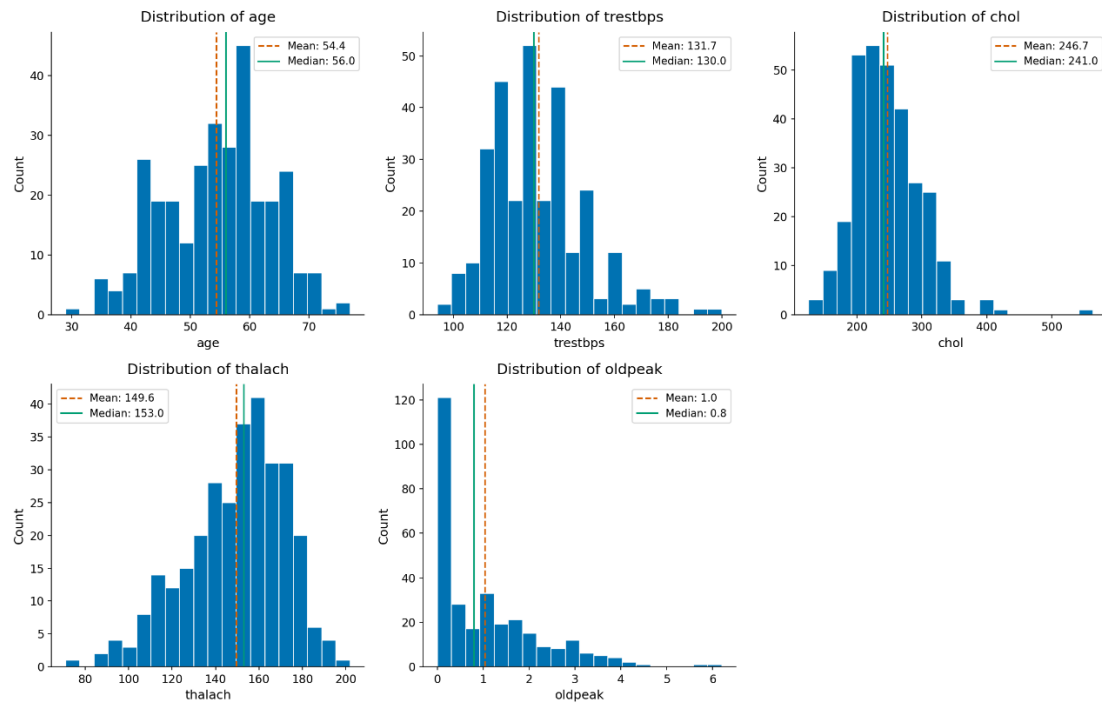
GitHub Repository for this project can be found here:

[https://github.com/mnooney/CVD\\_Capstone](https://github.com/mnooney/CVD_Capstone)

# Appendix

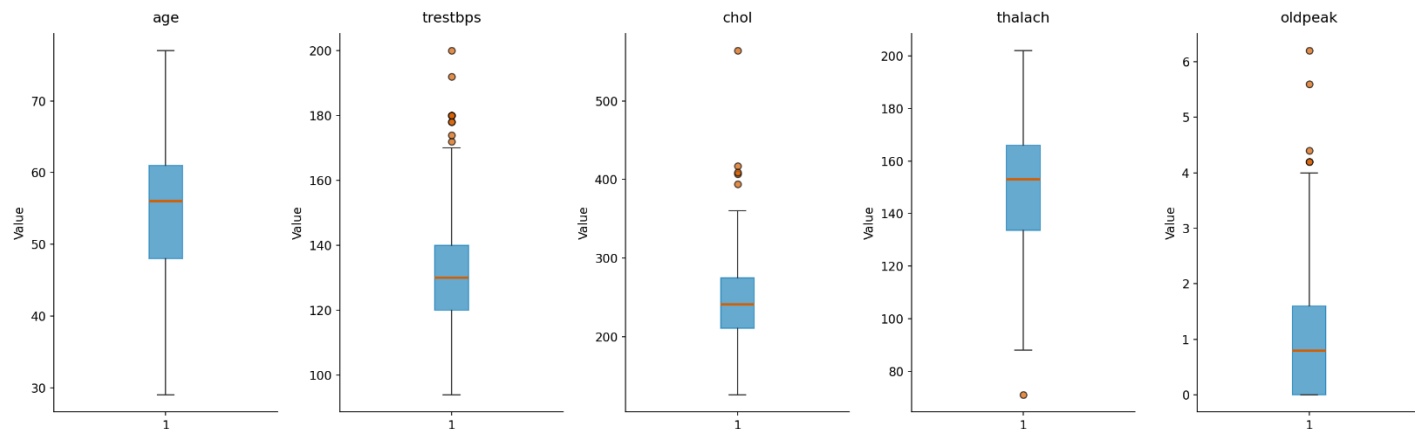
## Figures

**Figure 1: Distribution of Continuous Variables**

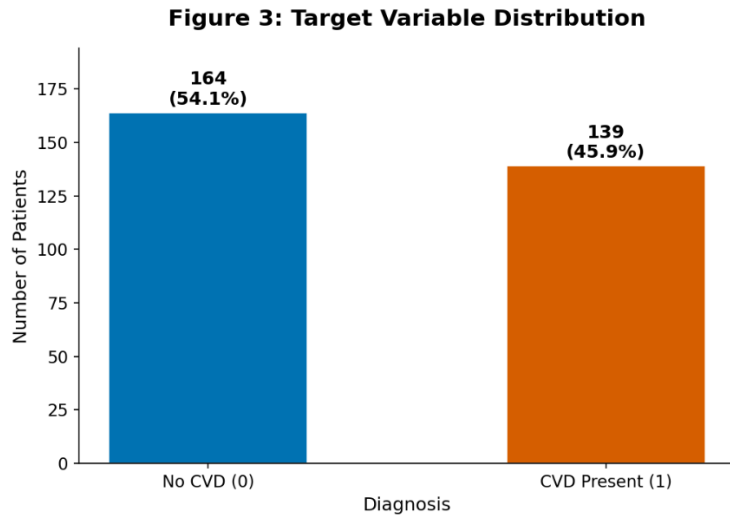


**Figure 1:** Histograms for all five continuous variables. Orange dashed lines indicate mean values. Green solid lines indicate median values. Right skew observed in cholesterol and oldpeak, with mean pulled right by high value outliers. Age and thalach show more symmetric distributions.

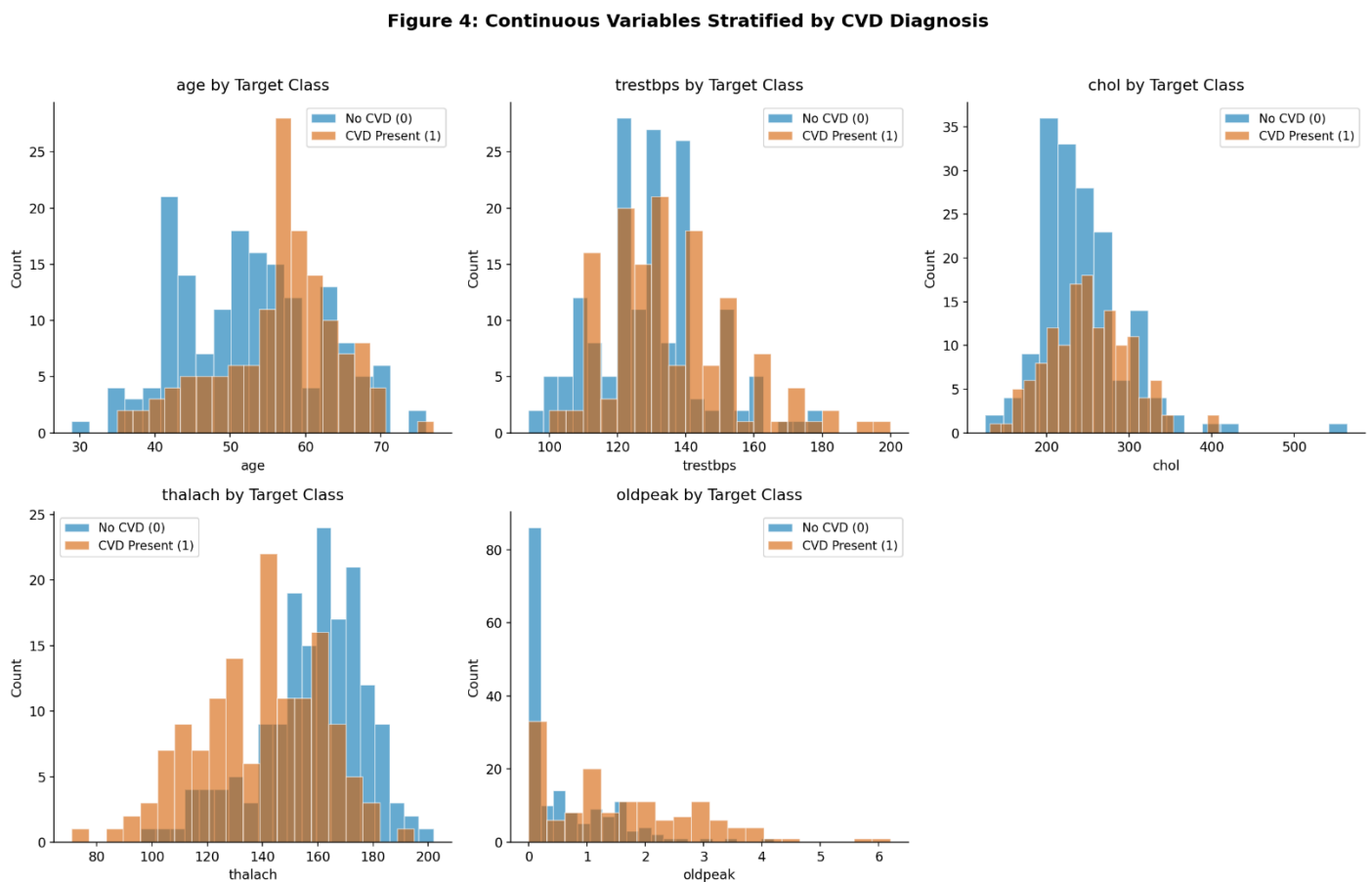
**Figure 2: Boxplots for Outlier Detection - Continuous Variables**



**Figure 2:** Boxplots for all five continuous variables. Orange horizontal line = median values. Orange dots = outliers. Cholesterol is showing an extreme outlier. Oldpeak shows multiple extreme outliers consistent with right skew identified in Figure 1. Resting blood pressure shows many outliers consistent with hypertensive patients. Age and maximum heart rate show minimal outliers.

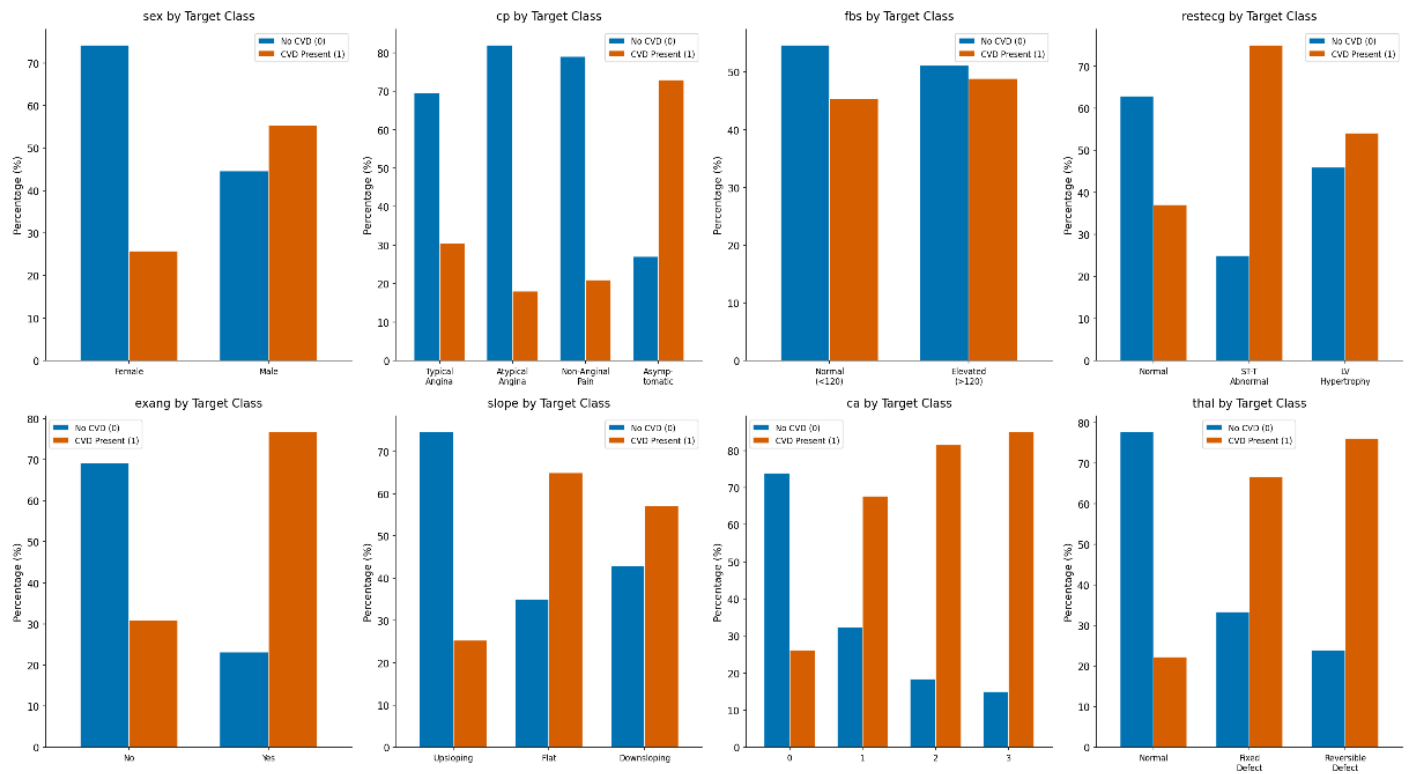


**Figure 3:** Distribution of binary target variable. Blue = No CVD (n = 164, 54.1%). Orange = CVD present (n = 139, 45.9%). Mild class imbalance shown.



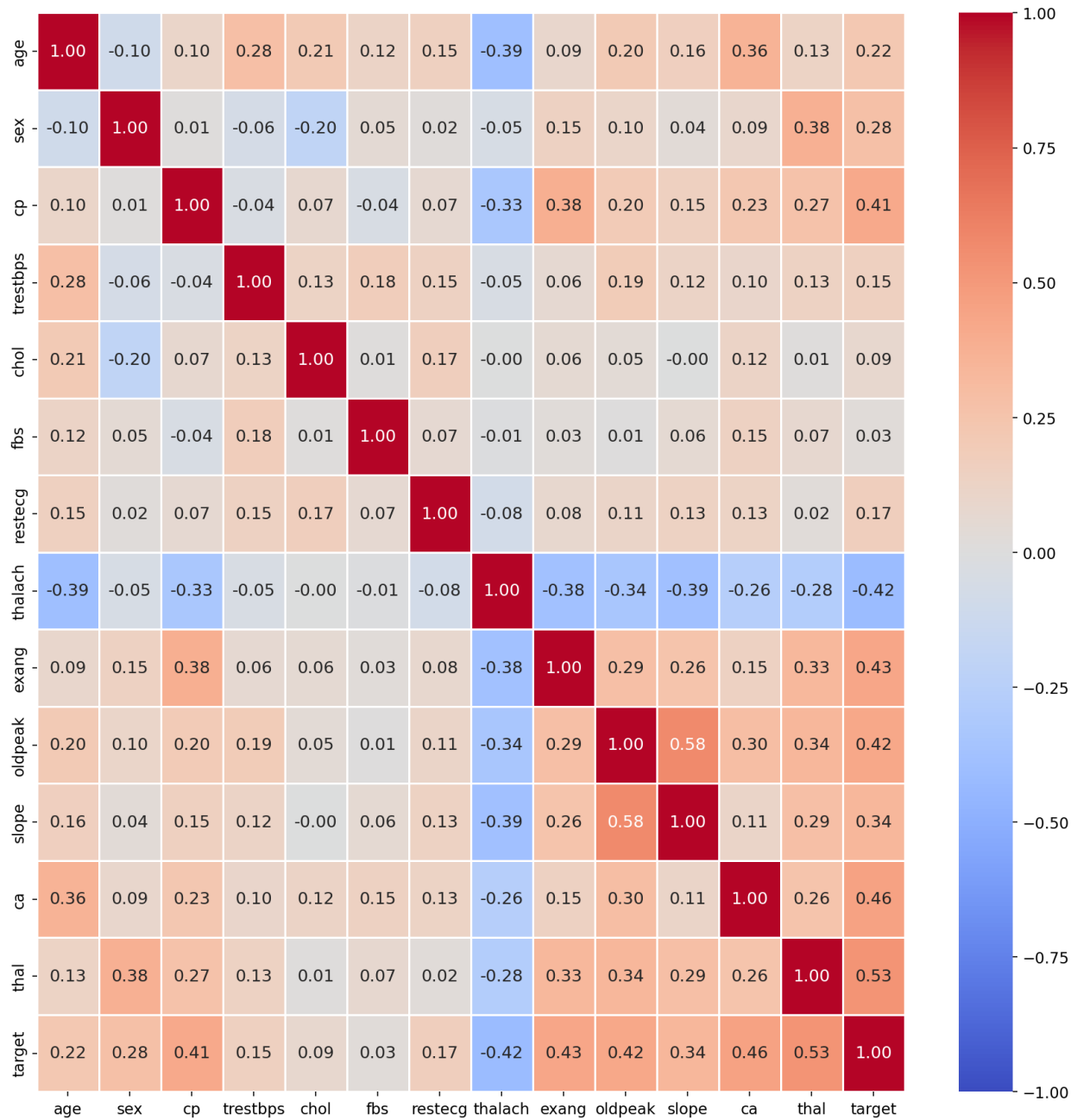
**Figure 4:** Overlapping histograms of continuous variables stratified by CVD diagnosis. Blue = no CVD. Orange = CVD present. Maximum heart rate (thalach) and ST depression (oldpeak) show clearest separation between classes suggesting a possible stronger predictor value. Cholesterol and resting blood pressure show most overlap between classes.

**Figure 5: Categorical Variables by CVD Diagnosis (%)**



**Figure 5:** Grouped bar charts showing percentage distribution of CVD diagnosis within each category of 8 categorical predictor variables. Blue = No CVD. Orange = CVD present. Number of blocked vessels (ca), exercise induced angina (Exang), chest pain type (cp), and thalassemia type (thal), show strongest associations with CVD diagnosis.

**Figure 6: Correlation Matrix**



**Figure 6:** Correlation heatmap of all variables. Red = positive correlation. Blue = negative correlation. Maximum heart rate (thalach), number of blocked vessels (ca), and thalassemia type (thal) show strongest correlations with target variable at -0.42, 0.46, and 0.53, respectively. Oldpeak and slope show the highest possible collinearity at 0.58.

## Tables

|                 | Mean   | SD    | Min   | Max   | Skewness | Kurtosis |
|-----------------|--------|-------|-------|-------|----------|----------|
| <b>age</b>      | 54.44  | 9.04  | 29.0  | 77.0  | -0.21    | -0.52    |
| <b>trestbps</b> | 131.69 | 17.60 | 94.0  | 200.0 | 0.71     | 0.88     |
| <b>chol</b>     | 246.69 | 51.78 | 126.0 | 564.0 | 1.14     | 4.49     |
| <b>thalach</b>  | 149.61 | 22.88 | 71.0  | 202.0 | -0.54    | -0.05    |
| <b>oldpeak</b>  | 1.04   | 1.16  | 0.0   | 6.2   | 1.27     | 1.58     |

**Table 1:** Continuous Variable Descriptive Statistics

|           | Variable | Category | Count | Proportion |
|-----------|----------|----------|-------|------------|
| <b>0</b>  | sex      | 0.0      | 97    | 0.320      |
| <b>1</b>  | sex      | 1.0      | 206   | 0.680      |
| <b>2</b>  | cp       | 1.0      | 23    | 0.076      |
| <b>3</b>  | cp       | 2.0      | 50    | 0.165      |
| <b>4</b>  | cp       | 3.0      | 86    | 0.284      |
| <b>5</b>  | cp       | 4.0      | 144   | 0.475      |
| <b>6</b>  | fbs      | 0.0      | 258   | 0.851      |
| <b>7</b>  | fbs      | 1.0      | 45    | 0.149      |
| <b>8</b>  | restecg  | 0.0      | 151   | 0.498      |
| <b>9</b>  | restecg  | 1.0      | 4     | 0.013      |
| <b>10</b> | restecg  | 2.0      | 148   | 0.488      |
| <b>11</b> | exang    | 0.0      | 204   | 0.673      |
| <b>12</b> | exang    | 1.0      | 99    | 0.327      |
| <b>13</b> | slope    | 1.0      | 142   | 0.469      |
| <b>14</b> | slope    | 2.0      | 140   | 0.462      |
| <b>15</b> | slope    | 3.0      | 21    | 0.069      |
| <b>16</b> | ca       | 0.0      | 176   | 0.581      |
| <b>17</b> | ca       | 1.0      | 65    | 0.215      |
| <b>18</b> | ca       | 2.0      | 38    | 0.125      |
| <b>19</b> | ca       | 3.0      | 20    | 0.066      |
| <b>20</b> | ca       | NaN      | 4     | 0.013      |
| <b>21</b> | thal     | 3.0      | 166   | 0.548      |
| <b>22</b> | thal     | 6.0      | 18    | 0.059      |
| <b>23</b> | thal     | 7.0      | 117   | 0.386      |
| <b>24</b> | thal     | NaN      | 2     | 0.007      |

**Table 2:** Categorical Variable Descriptive Statistics

## Data Dictionary

| Variable | Type          | Description                                        | Values/Range                                                                    |
|----------|---------------|----------------------------------------------------|---------------------------------------------------------------------------------|
| age      | Continuous    | Age of patient in years                            | 29 - 77                                                                         |
| sex      | Categorical   | Biological sex                                     | 0 = Female, 1 = Male                                                            |
| cp       | Categorical   | Chest pain type                                    | 1 = Typical Angina, 2 = Atypical Angina, 3 = Non-Anginal Pain, 4 = Asymptomatic |
| trestbps | Continuous    | Resting blood pressure on admission (mmHg)         | 94 - 200                                                                        |
| chol     | Continuous    | Serum cholesterol (mg/dl)                          | 126 - 564                                                                       |
| fbs      | Categorical   | Fasting blood sugar > 120 mg/dl                    | 0 = Normal, 1 = Elevated                                                        |
| restecg  | Categorical   | Resting electrocardiographic results               | 0 = Normal, 1 = ST-T Abnormality, 2 = LV Hypertrophy                            |
| thalach  | Continuous    | Maximum heart rate achieved during exercise (bpm)  | 71 - 202                                                                        |
| exang    | Categorical   | Exercise induced angina                            | 0 = No, 1 = Yes                                                                 |
| oldpeak  | Continuous    | ST depression induced by exercise relative to rest | 0.0 - 6.2                                                                       |
| slope    | Categorical   | Slope of peak exercise ST segment                  | 1 = Upsloping, 2 = Flat, 3 = Downsloping                                        |
| ca       | Categorical   | Number of major vessels colored by fluoroscopy     | 0 - 3, NaN = Missing                                                            |
| thal     | Categorical   | Thalassemia type                                   | 3 = Normal, 6 = Fixed Defect, 7 = Reversible Defect, NaN = Missing              |
| target   | Binary Target | Cardiovascular disease diagnosis                   | 0 = No CVD, 1 = CVD Present                                                     |

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